

Practice Quiz: Perception — Reading Curvature in the Fabric

Part A: Multiple Choice

1. A crocheted circle that cups into a bowl shape has: A) Zero Gaussian curvature — it is flat B) Negative Gaussian curvature — it has too many stitches C) Positive Gaussian curvature — it has too few stitches for its radius D) No curvature — it simply needs blocking
2. In a convolutional neural network, the first layer typically detects: A) Complete objects like faces or animals B) Simple local features like edges and transitions C) The overall meaning of the image D) Only color information, ignoring shape
3. The crocheter's four-level perceptual hierarchy (stitch, row, section, shape) is best compared to: A) A single-layer neural network with no hidden layers B) The hierarchical feature extraction in a deep convolutional neural network C) A lookup table with no processing D) Random noise that carries no information
4. When a crocheter feels that a fabric is ruffling when it should lie flat, this is an example of: A) A well-matched prediction — everything is as expected B) Topological prediction error — the sensory evidence contradicts the generative model's curvature prediction C) The absence of any generative model D) Zero free energy in the system
5. In Active Inference, top-down predictions in crochet perception means that: A) The crocheter only processes sensory data from the bottom up, never making predictions B) The crocheter's belief about the finished object shapes what they expect to feel and see at each stitch C) Predictions only flow from stitches to rows, never from the overall shape downward D) Top-down means looking at the project from above rather than from the side
6. Blocking a finished crochet piece is analogous to: A) Deleting the neural network and starting over B) Gradient descent — iteratively adjusting the surface to minimize the

difference between actual and target shape C) Increasing the learning rate to infinity D) Ignoring all prediction errors

7. An experienced crocheter spots a curvature error that a beginner misses because: A) The experienced crocheter has a lower-precision generative model B) The experienced crocheter's generative model generates more precise topological predictions, making mismatches more salient C) Beginners have better eyesight D) Experience makes crocheters less attentive to their work

Part B: Short Answer

1. A flat crochet circle in single crochet requires six increases per round. Explain what happens to the fabric's topology if you use eight increases per round instead, and describe how a crocheter would perceive this topological change through both touch and vision.
2. Compare the way a CNN builds feature representations across its layers to the way a crocheter builds perceptual understanding from individual stitch tension to overall project shape. Where does the analogy hold, and what is one important difference between the two systems?
3. You are crocheting a flat rectangular blanket and you notice the edges are starting to curve inward — the blanket is becoming narrower than it should be. Describe this situation as a topological prediction error. What does your generative model predict? What is the sensory evidence? What correction would you make, and how does this parallel how a neural network updates its weights after a loss signal?